🫁 Breathing, Exercise & Energy: Reading · ANSWER KEY

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

How Cells Get Energy from Food

Your cells need energy all the time. They get it through cellular respiration: glucose + oxygen → carbon dioxide + water + ATP. ATP is the energy molecule cells can use right away, like energy coins. Energy is released when a phosphate group comes off ATP. Cellular respiration happens in the mitochondria, and both plants and animals do it.

The lungs take in oxygen and get rid of carbon dioxide. The heart pumps blood that carries oxygen and glucose to every cell. When you exercise, your muscle cells need more ATP, so breathing rate and heart rate go up.

In the exercise lab, students measured three clues of respiration at rest and after two minutes of exercise: breathing rate, heart rate and carbon dioxide. Students breathed through a straw into BTB. After exercise, the BTB turned yellow faster, showing more CO2.

Combustion (burning) is the same reaction as respiration, only fast: fuel + oxygen → carbon dioxide + water + heat and light. A burning marshmallow loses mass because its atoms go into the air as CO2 and water. Our cells do the same thing slowly and safely.

Yeast is a one-celled fungus. In the Yeast Feast lab, yeast was given glucose, starch or water. The more CO2 bubbles, the more the yeast used that food. Yeast used glucose fastest because it is already the simple sugar cells use.

Check your reading

1. Write the word equation for cellular respiration.
Glucose + oxygen → carbon dioxide + water + ATP (energy).

2. Why do heart rate and breathing go up during exercise?
Muscle cells need more ATP, so they need more oxygen and glucose and make more CO2.

3. How are combustion and respiration alike?
Both use fuel + oxygen and give off CO2, water and energy.